



Twitter password crisis

Twitter had a password recovery crisis as some of its third-party apps caused 10000 user passwords to be revealed. <http://dgit.in/TwitterRec>

Data Wars

Another incident of cyber criminalism of US schemed recorded in Iran where the air defense, communication and power were targeted. <http://dgit.in/IranAttack>

Geek life

quantum chromodynamics is carried on by the system. The theory of study of elementary particles such as quarks and gluons is called as Quantum Chromodynamics (QCD). The System is used for research of quark-boson a phase of matter that holds the mystery of the creation of our universe.

IIT Delhi HPC

IIT Delhi has one of the fastest supercomputer in their campus. This system is a GPU centric high performance system and is one of the fewest in the world. NVIDIA experts worked with IIT Delhi team to built this system that is based on GPU Tesla Platform. IITD's system is ranked at 166th position in the top 500 list.

Specs:

System specification ae HP ProLiant XL230a server & HP ProLiant XL250a Gen9 server with NVIDIA K40M GPU cards. Total storage of the system is 1.5 PB. For network connections Infiniband interconnect technology is used. There are



NVIDIA Tesla powering IITD's supercomputer

in total 322 NVIDIA Tesla K40M GPU cards in the system. System performance is rated at 524.4 TFLOPS. CUDA and OpenACC are preferred programming platforms for the system.

This supercomputer is under the use of researchers in fields of biology, nano systems, atmospheric science and bioinformatics. System provides services across the IIT campus for study and research in the field of Data Analytics, Deep Learning, Computational Physics, Chemistry, Computational Fluid Dynamics and Material Science.

YOUR PC VS. INDIA'S FASTEST SUPERCOMPUTER

Let's see how your PC, laptop or phone matches up to India's fastest supercomputer. Here, we tell you how many of a device you'd need to equal the SahasraT (specs of the SahasraT in brackets). The devices we've chosen are the Nexus 6P, Asus G751J-DB7 (G-Sync) laptop, and an assembled ultra high-end gaming rig:

Core Count (33,024 cores)

- 4128 x Nexus 6P
- 8256 x ASUS G-Sync laptop
- 8256 x Ultra High-end Gaming RIG

RAM (5600 GB)

- 1866 x Nexus 6P
- 233 x ASUS G-Sync laptop
- 350 x Ultra High-end Gaming RIG

Storage (2.1 PB)

- 31,250 x Nexus 6P
- 1773 x ASUS G-Sync laptop
- 1593 x Ultra High-end Gaming RIG

Graphics Memory

- 132 x ASUS G-Sync laptop
- 88 x Ultra High-end Gaming RIG

Param Yuva 2

Param Yuva 2 is the supercomputer located at Center for Development of Advanced Computing (C-DAC) in Pune. The System is developed by Intel as OEM and system integrator is NetWeb technologies. Param Yuva 2 is ranked 251st in the world.

Specs:

Param Yuva 2 has 221 Intel Xeon E5-2670 nodes that also consist of Intel Xeon Phi 5110P, with Linux 64-bit (CentOS 6.2) and a total storage of 200TB. The System is rated at 388.44 TFLOPS. There are in all 3536 CPU cores and 26520 coprocessor cores. The system is interconnected with Mellanox FDR Infiniband primary interconnect for interconnection of nodes.

Param Yuva 2 provides service in fields of bioinformatics, space, weather forecasting, seismic data analysis, aerospace engineering, scientific research and pharmaceutical development. The system can also be interconnected with Indian Institute of Technology and National



CDACs Param Yuva-2

Institute of Technology via National Knowledge Network, which is a multi-gigabit backbone network for high speed data transfer between some educational institutions in India.

What does the future hold?

India shows promise in the field of supercomputing. We're not really where we should be in terms of the global scale. We have only 11 supercomputers featured in the top 500, and only 1 in the top 100. It's obvious that we need to do better and we will. The government of India is working towards this and has initiated the National Supercomputing Mission. Under the mission, the Government of India empowers an ambitious target of installing more than 70 high-performance computing facilities in the country. These computers will be connected by the National Knowledge Network.

This new project will not only boost Scientific Research and Development, but also improve the quality of higher education in Science & Technology. We certainly hope this dream is realised. India certainly needs to get to the forefront of supercomputing, because as a growing economy and one that hopes to bring more equality and uplift people out of poverty, we desperately need improved data crunching. We need to be able to predict the weather more accurately, have early warning for natural disasters, discover new life-saving pharmaceutical drugs, design rockets, and much more...